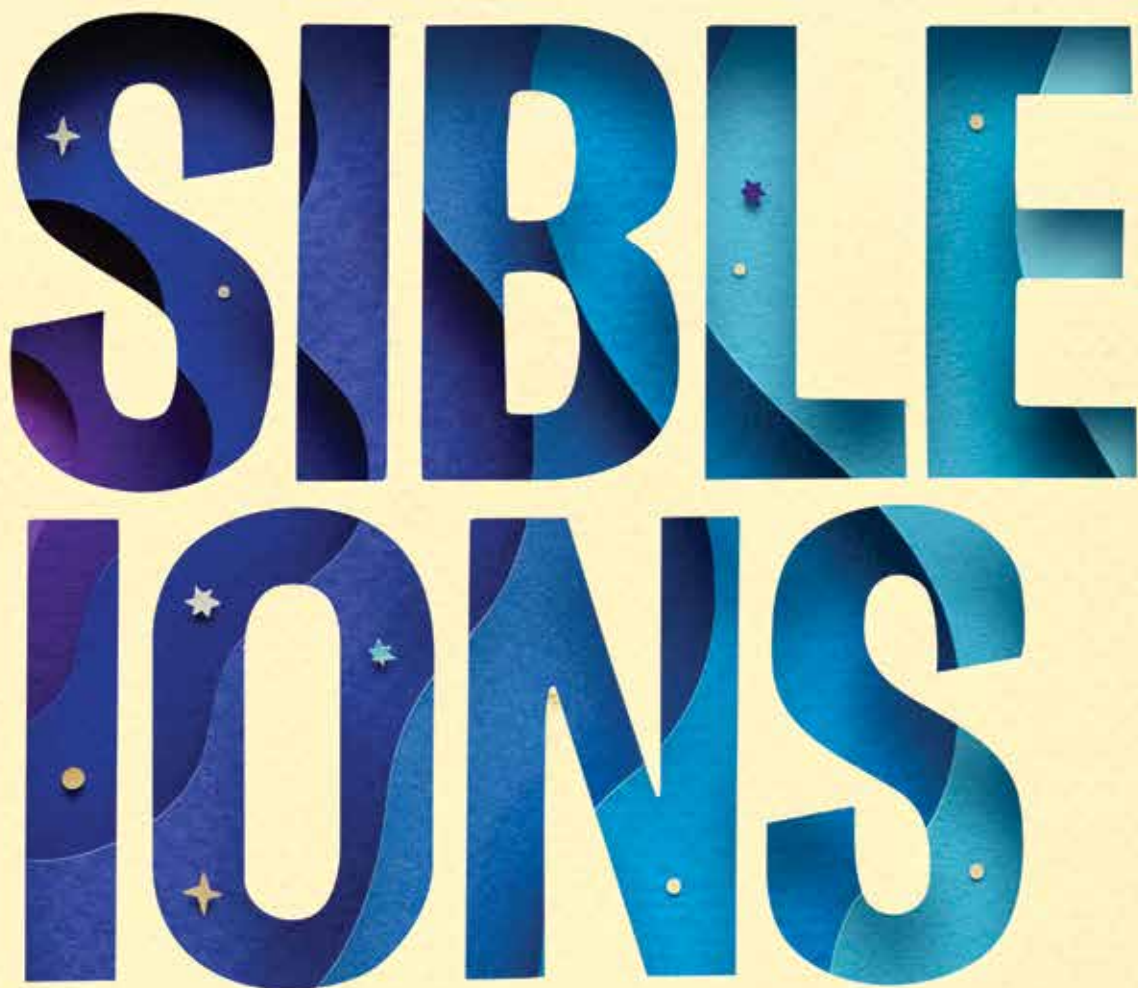
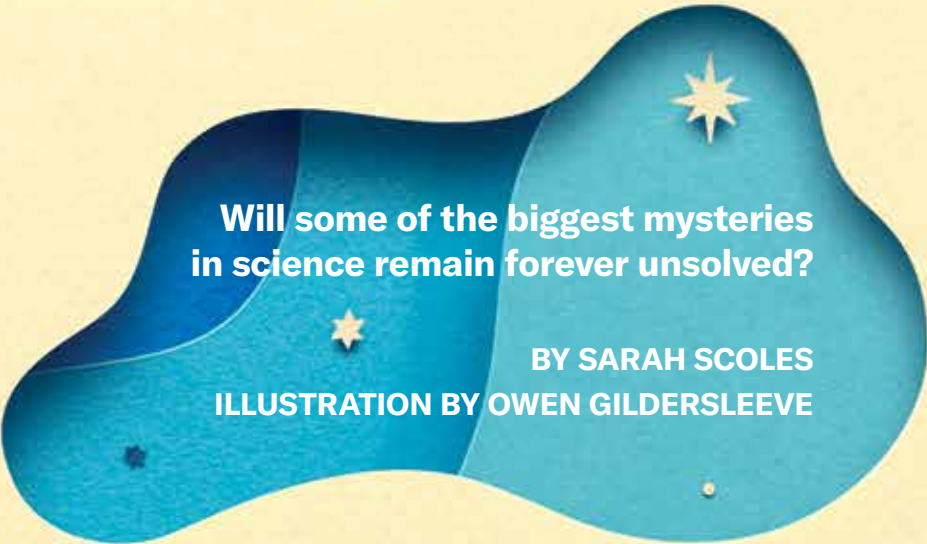


SPECIAL REPORT

IMPOSSIBLE QUEST



SIBBLE IONS



Will some of the biggest mysteries
in science remain forever unsolved?

BY SARAH SCOLES
ILLUSTRATION BY OWEN GILDERSLEEVE



SCIENCE, AT ITS BEST, IS A STRIVING TOWARD TRUTH. It is not, although we often think of it as such, the search for absolute truth. It's more a process of systematically gathering and interpreting data about the universe to understand its past and present and predict its future. What results is the best approximation we can access at any given time with the information, methods and instruments we have.

And what counts as the best approximation of truth often changes. Isaac Newton began to devise his apple-falling theory of gravity in 1666. It worked well for everyone's purposes in the 17th century. Then about 250 years later Albert Ein-

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stein stepped in with a better approximation: the general theory of relativity. Today some scientists are working to find ways to unite relativity with quantum mechanics, seeking a new approximation of the "truth" of gravity.

But are there some questions science can *never* fully answer? There are certainly questions scientists can't answer *yet*. There are also queries scientists once thought impossible to resolve that weren't. In 1835, for instance, philosopher Auguste Comte asserted that humans could never know what stars are made of. Then came spectroscopy, which allowed astronomers to parse stellar light for fingerprints of different chemicals.

Most things that seem forever unknowable eventually fall out of that porous basket. New ways of thinking or measuring expand the boundaries of science. This idea is the thesis of scholar

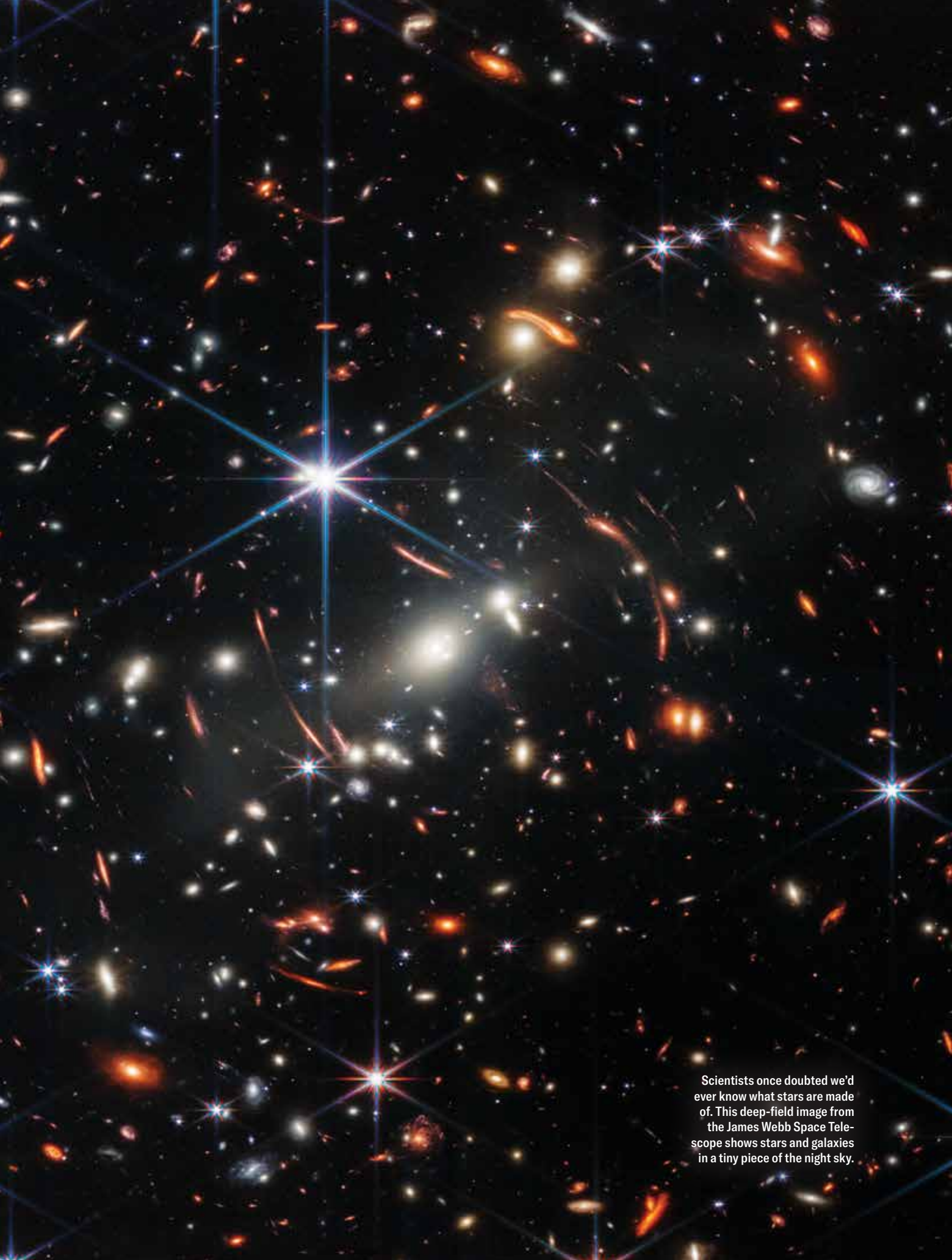
Alexander Krauss's most recent book, *The Engine of Scientific Discovery*. Until radio antennas existed, for instance, astronomers didn't know the universe emitted radio waves, so they didn't wonder what powered the gigantic jets shooting at nearly light speed from the centers of galaxies. Before microscopes, biologists didn't know about animalcules, as microbes were first called, so they didn't know they might someday study how the microbiome impacts mental health. "When scientists invent novel instruments and new computational and experimental techniques, they effectively expand the sensory and cognitive reach of science itself," Krauss says.

Still, not all questions will find answers. Some will likely remain beyond scientific grasp, even with methodological and interdisciplinary advances. Some will be so beyond our imagination that they will remain unknown unknowns forever. That's frustrating: science aims to understand the universe by observing, experimenting, interpreting, iterating. But that enterprise hits walls when there is nothing to observe or experiment on, when something lies past a fundamental barrier, or when observing or experimenting changes what is being observed or experimented on.

So how do we make sense of, and make peace with, necessary gaps in our knowledge? Perhaps by accepting that science isn't about finding the truth, the whole truth, and nothing but the truth. We'll just have a set of evolving best shots and, ideally, shots that keep leading—like a non-Brownian walk—to the next destination. ●

Sarah Scoles

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Scientists once doubted we'd ever know what stars are made of. This deep-field image from the James Webb Space Telescope shows stars and galaxies in a tiny piece of the night sky.